A TWO-STAGE CLASSIFICATION OF LINEAR GENERALIZED STONE–WALES PATHS IN GOLDBERG–COXETER IMAGES OF THE DODECAHEDRON

TATSUHIKO TAKIMOTO

Abstract. We study literal linear generalized Stone–Wales (gSW) paths in the duals of $\mathrm{GC}(k,l)[C_{20}]$, where $k > l > 0$. Such a path has distinct vertices, an even number of vertices, degree-five endpoints, degree-six neighbours of its endpoints, and triangular consecutive triples. Internal degree-five vertices are not excluded by the definition. We prove an explicit two-stage classification. Nonprimitive parameters give no path. For primitive parameters, put $m = \min(l, k-l)$. Apart from unit-width cases, existence is equivalent to $k \equiv \pm 1 \pmod{m}$ when $\lfloor k/m \rfloor \geq 3$, and to $m \equiv \pm 1 \pmod{r}$, where $r = k - 2m$, when that quotient is two. Eight periodic belts give positive witnesses; collisions preceding any admissible endpoint rule out the other cases. Consequences are a smallest primitive negative example $\mathrm{GC}(17,5)[C_{20}] = C_{7980}$ and eventual periodicity along integer affine rays. A density-one statement for negative primitive parameter points remains a separate proof draft. The complete classification and the primitive minimum are now proved in Lean on an independently specified closed surface graph. Comparator checks the full classification against the independent statement; six kernels accepted the same proof. Eventual periodicity is also formalized, while an executable threshold and period are not. The owner reports confirming the meaning of the independent challenge statement; human review of the proof remains outstanding. An appendix records a separate finite census of IPR fullerenes from C_{60} to C_{200}: 120,622,224 outputs from each of two generators, with 816 negative isomorphism classes. We distinguish finite checks, AI prose review, formal proofs, and independently constructed verification. Novelty relative to the full position-mapping and intersection theory remains unsettled.

Editorial status. This public-release edition v8.1 continues English v7.2 and Japanese v4. The work is AI-assisted. The theorem statements below retain v4's numbering; calling a statement a theorem does not assert completed external peer review. The coordinator (Claude) reviewed the Japanese v4 text of Appendices B and C at 03:46 JST on 19 October 2026 (commit `49430e1c`), recalculating its formulas and recording no missing steps. This English text transcribes that Japanese argument; selected English formulas were also machine-checked by the coordinator. That historical review was AI prose review. The later formal proof checks the exact graph statement described in Section 9; it does not replace human review. P006-15 separately records the owner's reported confirmation of the challenge statement's meaning on 19 October 2026; this is not proof peer review. The coordinator's later addendum also completes AI prose review of A-sheet items 2 and 3; Section 11 records the scope. The author has approved the authorship, affiliation, licenses and AI-use statement for this edition. The separate treatment of other families and the general ray theorem is reserved for a follow-up paper.

1. Objects and definitions

1.1. **The literal path definition.** A fullerene is a simple, three-connected, cubic plane graph all of whose faces are pentagons or hexagons. Its dual is a spherical triangulation with twelve degree-five vertices and all other vertices of degree six. Following Definition 1(a) of [1], a literal gSW path is a sequence $p_0, \ldots, p_e$ satisfying:

(1) $e \geq 3$ is odd and all vertices are distinct;
(2) p_i is adjacent to p_{i+1} and to p_{i+2} whenever defined;

Date: Version 8.1, 20 October 2026.

(3) p_0, p_e have degree five, and p_1, p_{e-1} have degree six.

The source's notation $v_1, \dots, v_{2w}$, $w \geq 2$, gives $e = 2w - 1$. Consecutive adjacency is part of the word "path"; distance-two adjacency is explicit. In particular, internal degree-five vertices are allowed. Requiring an even number of vertices is a definition, not a new necessary condition. "Positive" and "negative" mean presence and absence of this path. Reaction barriers, radial gSW operations, and operations permitting heptagons are outside our scope.

1.2. **The A3 family and lattice coordinates.** Write $A3(k,l) = \mathrm{GC}(k,l)[C_{20}]$, $k > l > 0$. Graver's older coordinates $A3(p,q)$ correspond to $(k,l) = (p+q, p)$. Develop each parent face of the icosahedral dual in the triangular lattice with

$$O = (0,0), \quad e = (k,l), \quad f = (-l, k+l), \quad T = k^2 + kl + l^2.$$

The fullerene has $20T$ vertices and its dual has $10T + 2$. Let $R(x,y) = (-y, x+y)$ and $G(u,v) = ue + vf$. The maps G and R commute, and the inverse coordinate change is

$$u = \frac{(k+l)x + ly}{T}, \qquad v = \frac{-lx + ky}{T}. \tag{1}$$

In parent coordinates write

$$U(i,j) = [(i,j), (i+1,j), (i,j+1)], \quad D(i,j) = [(i+1,j), (i+1,j+1), (i,j+1)].$$

Developments use integer powers of R and integer translations. A fine unit triangle need not lie in one parent face: pieces cut by parent edges are mapped separately. If $\gcd(k,l) = 1$, the vectors $e, f, f-e$ are primitive, so no fine lattice vertex lies inside a parent edge. Identifications of ordinary vertices can therefore be tested in face interiors; parent vertices require separate treatment. Distinct face labels do not imply disjoint closures.

1.3. **Kinds of evidence.** An AI proof draft is a general argument awaiting review. AI prose review means the specified formulas or paragraphs were read and rederived by Claude; it is neither human peer review nor necessarily an independent derivation. Lean checks the stated formal propositions, separately from the geometric hypotheses. Finite checks on an independent construction are identified by both the constructor and the checker. "R004 independent" refers to completed reports from a verification team that did not read the main team's proof or implementation within the stated scope. Two independently derived general prose proofs are now received: the research team's argument and R004's alternative. The team's formalization, completed with the coordinator's positive construction, proves the complete classification on the independent graph (P006-14). This is not a third independent derivation. Sections 9 and 11 give the precise boundary.

2. The classification

Theorem 1 (Two-stage classification)**.** *Let $k > l > 0$ and $d = \gcd(k,l)$. If $d > 1$, then $A3(k,l)$ is negative. If $d = 1$, put $m = \min(l, k-l)$. The graph is positive if and only if one of the following holds:*

$$\begin{aligned} &m = 1; \\ &m \geq 2, \quad q = \lfloor k/m \rfloor \geq 3, \quad k \equiv \pm 1 \pmod{m}; \\ &m \geq 2, \quad q = 2, \quad r = k - 2m, \quad [\, r = 1 \text{ or } m \equiv \pm 1 \pmod{r} \,]. \end{aligned} \tag{2}$$

Machine-readable claim: `thm:classification`.

For primitive parameters with $m \geq 2$, equality $k = 2m$ is impossible; thus $0 < r < m$ in the last line. No congruence modulo zero is used. We handle $r = 1$ before the remaining congruences.

The proof separates the integer-inflation obstruction (Section 3), the two start orbits (Section 4), positive paths (Section 5), and collisions with endpoint exclusion (Sections 6–7). The four regions in the original coordinates are as follows.

Region	Positive construction	Negative construction
$k > 3l$	AR vertical/horizontal belts	Outer fan
$2l < k < 3l$	AT second belt	Positive fan, $c = 2$
$l < k < 2l$, $k \geq 3(k-l)$	Diagonal belt	Positive fan, $c = 1$
$l < k < 2l$, $k < 3(k-l)$	Folded second belt	Negative fan

Primitive points with $k = 2l$ or $3l$ have $l = 1$ and belong to Section 5.1. In the third region, $k \bmod (k-l) = l \bmod (k-l)$. In the fourth, $r = 2l-k$ and $m = k-l = l-r$, so $m \bmod r = l \bmod r$. This gives (2), with the assembly reviewed in AU, Section 7. The equality of decisions at (k, l) and $(k, k-l)$ follows from the classification, not from a graph isomorphism: their norms differ by $2k(k-2l)$ and their orders generally differ.

3. The obstruction for nonprimitive parameters

Machine-readable claim: `sec:inflation`.

If $d = \gcd(k, l) > 1$, the construction is the integer d-fold inflation of $\mathrm{GC}(k/d, l/d)[C_{20}]$. More generally, in $\mathrm{GC}(d, 0)[F]$ the degree-five parent points lie in the d-multiple lattice in each chart. Develop the path from its initial triangle. At an odd position, the determinant of the lattice point with the direction of progression is ± 1. If this point were degree five, that determinant would have to be divisible by d, a contradiction. The gluings are lattice rotations through multiples of 60 degrees and transfers about points of the d-multiple lattice, preserving divisibility. This does not assume a global prohibition on internal degree-five vertices.

The nonprimitive absence statement is now proved on the full closed quotient, including lattice points on parent edges, and transported by a proved graph isomorphism to the independent specification. It uses actual degrees and does not forbid internal degree-five vertices. R004 also supplies an independently derived prose argument. The original argument is `general-inflation-argument.md`.

4. Two starts and candidate endpoints

Machine-readable claim: `sec:starts`.

4.1. **Forced continuation and the first repeat.** Suppose every edge has exactly two common neighbours. From an oriented triangle (a, b, c), the next vertex is the common neighbour of b, c other than a. Every simple path is a prefix of this forced walk. If ρ is its first repetition index, existence is equivalent to an admissible endpoint before ρ, and $\rho \leq |V|$. This conditional principle is formalized as `R002.finite_literal_path_decision`. The local assumption and all-start coverage remain hypotheses of that conditional theorem. The later full classification additionally derives the surface degrees, common-neighbour property and start reduction in its own concrete graph modules. The noncomputable `forcedWalk` is not a certified executable classifier; see `formal-correspondence-decision.md`.

4.2. **The two rotational orbits.** The 60 orientation-preserving parent rotations act on the refined lattice. Cyclic permutations of a face's corners act locally by $R^{2p}z + (0, e, f)_p$ and respect the edge gluings. They are transitive on the twelve parent vertices, with the fivefold stabilizer transitive on the five initial radial edges. The two choices of third vertex have opposite orientation and are not exchanged. Thus the 120 starts split into two orbits of size 60, represented by

$$A = ((0,0),(1,0),(0,1)), \qquad B = ((0,0),(0,1),(1,0)).$$

Their developments are

$$\begin{aligned} A: &\quad P_{2j} = (0, j), \quad P_{2j+1} = (1, j), \\ B: &\quad P_{2j} = (j, 0), \quad P_{2j+1} = (j, 1). \end{aligned} \tag{3}$$

The AQ two-orbit argument was reviewed in prose. A negative proof must exclude both representatives.

4.3. **The first parent-lattice candidates.** Assume $\gcd(k,l)=1$ and $l\geq 2$. Since k,l are coprime to T, there is a unique $0<h<T$ with $kh\equiv l \pmod T$. The first odd parent-lattice candidates in (1) are

$$A:(1,h),\ E_A=2h+1; \qquad B:(t_B,1),\ t_B=T-h-1,\ E_B=2t_B+1. \tag{4}$$

The first positive even candidate has index $2T$. The integer parent coordinates u_A, v_B satisfy

$$\begin{aligned} Tu_A &= k+l+lh, \qquad Tv_B = k-lt_B, \\ ku_A &\equiv 1 \pmod l, \quad k(-v_B)\equiv -1 \pmod l. \end{aligned} \tag{5}$$

These formulas apply before a cone is encountered. They do not prove that the candidates are reachable without repetition. Positive cases require simplicity of the entire prefix; negative cases require a separate exclusion of earlier parent vertices.

5. Positive cases: eight belts and their caps

Machine-readable claim: `sec:positive`.

5.1. **The boundary $l=1$.** Take the B sequence through $P_{2k+1}=(k,1)=e$. Its rectangle $0\leq x\leq k$, $0\leq y\leq 1$ lies in $U(0,0)\cup D(0,-1)$, whose face labels differ. Each face map is injective and no fine vertex lies inside the common parent edge. Only the endpoints are parent vertices, giving an admissible path.

5.2. **Finite belt tables and infinite developments.** Let F_z be the map from a developed parent triangle to its quotient face. A period v_0 means $F_{z+v_0}(x)=F_z(x-v_0)$ on corresponding cells: the affine map, not just the face label, repeats. Equality one period apart and unique reversible edge gluing extend the table to all integer cells.

Belt	Parent strip	v_0	Faces per period
AR A	$0\leq u\leq 1$	$(0,5)$	10
AR B	$-1\leq v\leq 0$	$(5,0)$	10
AT A	$0\leq v-2u\leq 1$	$(2,4)$	12
AT B	$0\leq u+3v\leq 1$	$(6,-2)$	12
Diagonal A	$0\leq v-u\leq 1$	$(3,3)$	12
Diagonal B	$0\leq u+2v\leq 1$	$(6,-3)$	12
Folded second A	$-1\leq v-2u\leq 0$	$(2,4)$	12
Folded second B	$-1\leq u+3v\leq 0$	$(6,-2)$	12

AR A alternates $U(0,j), D(0,j)$; AR B alternates $D(j,-1), U(j+1,-1)$. A diagonal cell i uses $U(i,i), D(i,i), U(i,i+1), D(i,i+1)$. AT A uses $U(i,j), D(i,j)$ for $j=2i, 2i+1, 2i+2$, and folded second A uses $j=2i-1, 2i, 2i+1$. For each B belt rotate the triangle list by R^{-1} and reglue from its specified initial face; do not simply reuse rotated quotient labels.

The faces within a period are distinct, and the parent-vertex label sets on the two boundaries are disjoint. Additional cap faces are distinct from the central faces and from one another. There are 20 phase entries: $5+5$ for AR B, $1+2$ for AT A, $1+3$ for diagonal A, and $1+2$ for folded second B. Their vertices, rotations, and translations are recorded in `review-AW/cap-separation.md` and the fixed JSON tables. T0743 checked the eight belts and T0746 the 20 cap entries on Claude's independent construction. General containment is a separate step.

5.3. **Endpoints.** Use (3) up to index E in the following table. Unit-width overlaps are handled by a width-one or AR construction. If both signs coincide at width two, choose the plus sign; the minus cap arguments use width at least three.

Condition	Start	Q; endpoint index	Parent endpoint W
$l \geq 2$, $k = ql + 1$	A	$q(k+l)+l$; $2Q+1$	$(1, q)$
$l \geq 2$, $k = ql - 1$	B	$q(k+l)+l$; $2Q-1$	$(q+1, -1)$
$2l < k < 3l$, $r = k - 2l$, $l = sr + 1$	B	$s(3k+l)+k$; $2Q+1$	$(3s+1, -s)$
Same region, $l = sr - 1$	A	$s(2k+3l)+k+l$; $2Q+1$	$(s, 2s+1)$
$l < k < 2l$, $w = k - l$, $l = sw + 1$	B	$s(2k+l)+k$; $2Q+1$	$(2s+1, -s)$
Same region, $l = sw - 1$	A	$s(k+2l)+k+l$; $2Q+1$	$(s, s+1)$
$l < k < 2l$, $w = 2l - k$, $l = sw + 1$	A	$s(2k+3l)-k-l$; $2Q+1$	$(s, 2s-1)$
Same region, $l = sw - 1$	B	$s(3k+l)-k$; $2Q+1$	$(3s-1, -s)$

For AR, $kQ = qT + \varepsilon l$ and $lQ = T - \varepsilon(k+l)$. For the remaining six rows, respectively,

$$\begin{aligned} rQ + (3k+l) &= T, \quad rQ - (2k+3l) = T, \\ wQ + (2k+l) &= T, \quad wQ - (k+2l) = T, \\ wQ + (2k+3l) &= T, \quad wQ - (3k+l) = T. \end{aligned} \tag{6}$$

In each A row $G(W) = (1, Q)$, and in each B row $G(W) = (Q, 1)$, except AR B, where it is $(Q-1, 1)$. These integer identities are among the Lean lemmas in Section 9; bounds, simplicity, and legality are not consequences of the identities alone.

5.4. **Containment, including pieces cut by parent edges.** Use $[0,1] \times [0,Q]$ for A and $[0,Q] \times [0,1]$ for B, with the upper horizontal bound $Q-1$ for AR B. These rectangles contain all fine unit triangles used by the paths. Without caps, (1) and (6) put the normal coordinate in the strip interval; for example, second belt B has normal $[rx + (3k+l)y]/T \in [0,1]$.

For AR B, odd points above the strip have $t = 0, \ldots, q-1$ and even points below it have $t = Q-q, \ldots, Q-1$. The upper triangles have vertices with $t \leq q$. The bounds

$$T - k(q+1) - l = k(k-q-1) + kl + l^2 - l > 0, \quad k - q - 1 = q(l-1) - 2 \geq 0$$

put them in the initial face together with the belt. Point reflection $P_{E-j} = P_E - P_j$ covers the terminal cap. For second A, put $n = v - 2u$. If $n < 0$, then $x = ru - ln$ and $y \geq 0$ give $0 \leq u \leq 1/r$, $-u \leq v < 2u$, and $u + v < 3/r \leq 1$. This lies in $U(0,0) \cup D(0,-1)$. Reflection $z \mapsto W - z$ sends the $n > 1$ cap back to this initial piece and gives terminal faces $D(s-1, 2s), U(s-1, 2s+1)$. For diagonal A, use $n = v - u$, $x = wu - ln$ to obtain $0 \leq u \leq 1/w$, $-u \leq v < u$, $u + v < 2/w \leq 1$, and terminal faces $D(s-1, s), U(s-1, s+1)$.

For folded second B, $n = u + 3v$ and $y = ln - wv$. On the initial $n > 0$ piece,

$$u = \frac{3y - (3l-w)n}{w} < \frac{3}{w} \leq 1, \qquad u + v = \frac{2y - kn}{w} < \frac{2}{w} < 1.$$

Both are nonnegative, so the piece lies in $U(0,0) \cup D(0,-1)$; D is central and U additional. Reflection sends n to $-1-n$. These are planar containment arguments, not assertions of a reflection symmetry of the quotient surface. Appendix C gives the terminal pieces explicitly.

5.5. **Simplicity and degree conditions.** Two ordinary vertices identified in the central belt must have the same face phase, so their developed difference is $aG(v_0)$, $a \in \mathbb{Z}$. Across the width-one path rectangle the relevant transverse period magnitudes are $5l, 2r, 3w, 2w$ for AR, second, diagonal, and folded second belts. Each exceeds one, excluding $a \neq 0$; $a = 0$ identifies only the same index of (3). Cap labels are distinct and each face map is injective, excluding further repeats.

Every construction has $Q < T$, excluding positive even parent points. Substituting the possible integer normals $0, 1$ (or $-1, 0$) in (1) and (6) leaves just the odd endpoint. For diagonal B with $w = 1$, the possible normal-zero point $(-2, 1)$ is outside the rectangle. For folded second A with $w = 1$ it would have $u = -1$, again outside. Minus-sign caps use width at least three and contain

no other parent point. The endpoints belong to disjoint boundary-label sets. Thus the paths are simple, have even order, and meet all degree conditions.

6. Negative cases: collisions after leaving a belt

Machine-readable claim: `sec:negative`.

The Japanese v4 B-side argument has coordinator AI prose review (`49430e1c`). This English transcription retains the stated trust limitations in Section 11. The specified collision need not be the first repeat: an earlier repeat is also an obstruction. We must follow every fine triangle up to that time in the stated development and exclude an earlier legal endpoint. Section 7 supplies the latter exclusion.

6.1. The outer region.

Let $k = ql + r$, $q \geq 3$, $2 \leq r \leq l - 2$, and put

$$a = (q-4)k + (q+2)l - 1, \qquad b = (q+2)k + 6l - 1.$$

A leaves $D(0,q)$ for $U(1,q), D(1,q), U(1,q+1), D(1,q+1)$; the final two face maps compose the old belt map with $H_A(z) = Rz + (q+2,-1)$. B leaves $U(q+1,-1)$ for $D(q+1,-2), U(q+2,-2), D(q+2,-2), U(q+3,-2)$, using $H_B(z) = Rz + (q+1,-q-2)$ on the final two. The rotation centres are $(1,q+1)$ and $(q+2,-1)$. The finite table `AT-A3-collision-atlas.json` covers q mod 5. The relations

$$H_A(q+5, z+(0,5)) = H_A(q,z) + (0,5), \quad H_B(q+5, z+(5,0)) = H_B(q,z) + (5,0)$$

and belt periodicity extend it to all q.

The A rectangle $0 \leq x \leq 1$, $0 \leq y \leq b$ has $0 \leq u < 2$, $v < q+2$. It crosses $u = 1$ at $v = q + (r-x)/l \in (q, q+1)$. For $u \leq 1$ one has $v < q+1$; for $u \geq 1$, $v > q$. It is therefore covered by the belt and the four added faces. The last point $(0,b)$ has $1 < u < 2$, $q+1 < v < q+2$. For B, $-2 < v < 1$ and $u + v < q + 2$; its initial $v > 0$ piece lies in $U(0,0)$. At $v = -1$, $u = q + 1 + (r+y)/l \in (q+1, q+2)$, leading into the four lower faces. The last point $(b,1)$ has $-2 < v < -1$ and $q + 1 < u + v < q + 2$. Neither rectangle crosses the cut beyond the rotation centre. The following six positive quantities justify these claims:

$$\begin{aligned}
2T - k - l - lb &= (q^2-4)l^2 + 3qrl + 2r^2 - ql - r, \\
(q+2)T - kb &= (q-4)kl + (q+2)l^2 + k, \\
(q+2)T - kb - k - l &= (q-4)kl + (q+2)l^2 - l, \\
lb - T &= k(l-r) + 5l^2 - l, \\
kb - (q+1)T &= (4q-1)l^2 + (q+5)rl + r^2 - ql - r, \\
lb - T - k &= k(l-r-1) + 5l^2 - l.
\end{aligned}$$

For $q = 3$, substitute $k = 3l + r$, $r \leq l - 2$ in the second and third lines; for $q \geq 4$, positivity follows directly. The collision identities are

$$\begin{aligned}
(1,a) &= R(0,b) + (q+2)e - 6f, \\
(a,0) &= R(b,1) + (q-4)e - (q+2)f.
\end{aligned} \tag{7}$$

These differ from the maps H_A, H_B by the periods $-5f, -5e$. The early A point has $0 < u < 1$ and the early B point $-1 < v \leq 0$; for example $T - k - l - la = kb - (q+1)T > 0$. Since $0 \leq a < b$, the actual revisits are $2b \to 2a + 1$ for A and $2b + 1 \to 2a$ for B.

6.2. **The two positive-side fan regions.** Put $k = cl+w$, $l = sw+d$, $2 \le d \le w-2$, with $c = 1, 2$ and $s \ge 2, 1$, respectively. The strip $0 \le v - cu \le 1$ has period $p(1, c)$, where $p = 3, 2$, respectively. Set

$$a = (c+1)sk + (s-5)l - 1, \qquad b = (cs+5)k + (c+1)sl - 1,$$
$$W = (s+1, cs+c+1), \qquad H(z) = R^{-1}z + (-W_v, W_u + W_v).$$

From $D(s, W_v - 1)$ pass through $U(s, W_v), D(s, W_v), U(s+1, W_v)$. On the last two faces, the old belt map composed with H is the new map. B uses the rotated triangle list and translation, glued anew. The table `AU-A3-fan-atlas.json` covers every s mod p.

Rotate B's fine and parent coordinates by R only for comparison. In common fine coordinates (X, Y), A has $0 \le X \le 1$, $0 \le Y \le b$; B has $-1 \le X \le 0$, $0 \le X+Y \le b$. Since $X = wu - l(v - cu)$, the normal-one crossing is $u = s + (d + X)/w \in (s, s+1)$, before the cut from W. The bounds $v < W_v + 1$, $u + v < s + W_v + 2$ cover the portion beyond the strip by the original cell and the three added triangles. If $u \le s+1, v \le W_v$, use the original cell; if $v > W_v$, use the new faces; if $u > s+1$, use the final U. The initial negative-normal A piece has $u \le 1/w$, $-u \le v < cu$, $u + v < (c+1)/w \le 3/4$, and lies in $U(0,0) \cup D(0,-1)$. B has no such piece. The last points, A's $(1, b)$ and B's original $(b, 0)$, have $v > W_v$, $u + v > s + W_v + 1$.

The nine quantities needed for these bounds, the early point, and time ordering are strictly positive:

$$\begin{gathered} a, \quad b-a, \quad T-b, \\ (cs+c+2)T - (kb+k+l), \\ ((c+1)s+c+3)T - ((k+l)b+k), \\ kb - l - (cs+c+1)T, \\ (k+l)b - ((c+1)s+c+2)T, \\ T - [wa + (c+1)k + l], \qquad (s+2)T - (k+l+lb). \end{gathered} \tag{8}$$

Substitute $s = S + s_{\min}$, $w = D + E + 4$, $d = D + 2$, with $S, D, E \ge 0$, where $s_{\min} = 2, 1$. All 18 polynomials have nonnegative coefficients and positive constant terms. Every permitted parameter is obtained by $S = s - s_{\min}$, $D = d - 2$, $E = w - d - 2$. This is a universal coefficient proof, not a finite extrapolation. Lean checks the 18 inequalities, while their geometric interpretation is separate. The collision identities are

$$\begin{aligned} (0, a) &= R^{-1}(1, b) - (cs+5)e + (c+1)sf, \\ (a, 1) &= R^{-1}(b, 0) + (s-5)e + (cs+5)f. \end{aligned} \tag{9}$$

The images differ from H by one belt period. The eighth quantity puts the early point inside the old belt. Thus A revisits $2b + 1 \to 2a$, B revisits $2b \to 2a + 1$, and $0 < a < b < T$.

6.3. **The negative-side second region.** Assume $3l/2 < k < 2l$, $w = 2l - k$, $l = sw + d$, $s \ge 2$, $2 \le d \le w - 2$. The strip is $-1 \le v - 2u \le 0$, with period $(2, 4)$. Set

$$a = (2s-5)k + 3sl - 1, \qquad b = 3sk + (s+5)l - 1,$$
$$W = (s+1, 2s+1), \qquad H(z) = Rz + (3s+2, -s-1).$$

Leave $D(s, 2s)$ through $U(s+1, 2s), D(s+1, 2s), U(s+1, 2s+1)$. The last two maps use H. B uses the R^{-1}-rotated table. `AU-A3-negative-fan-atlas.json` covers s mod 2. Now $X = -wu - l(v - 2u)$, so at normal -1, $u = s + (d - X)/w \in (s, s+1)$. The eight positive quantities are

$$\begin{gathered} a, \quad b-a, \quad T-b, \quad (s+2)T - (k+l+lb), \\ (3s+3)T - ((k+l)b+k), \quad lb - k - (s+1)T, \\ (k+l)b - (3s+2)T, \quad T - (wa + 2k + 3l). \end{gathered} \tag{10}$$

The same nonnegative substitution, with $s = S + 2$, proves positivity and is formalized by eight Lean lemmas. It gives $u < s+2$, $u+v < 3s+3$ throughout the rectangle and $u > s+1$, $u+v > 3s+2$ at the last point. Outside the belt, $u \le s+1$ implies $v > 2s-1$ and $u > s+1$ implies $v > 2s$, using $k(s+1) - 1 - 2sl = (s-1)w + 2d - 1 > 0$ in the latter case. These bounds put every piece in the stated triangles without crossing the cut. A has no initial positive-normal cap; B's initial cap lies in the two faces of Section 5.4. Finally,

$$\begin{aligned}(1, a) &= R(0, b) + 3se - (s+5)f,\\ (a, 0) &= R(b, 1) + (2s-5)e - 3sf.\end{aligned} \tag{11}$$

The difference from H is one belt period, giving A's $2b \to 2a+1$ and B's $2b+1 \to 2a$ as actual ordinary-vertex repeats.

7. Endpoint ordering and assembly

Machine-readable claim: `sec:ordering`.

In the outer region, the nonunit residue and (5), with $u_A \ge 1$, $v_B < 1$, imply $u_A \ge 2$, $v_B \le -2$. Hence

$$lh = Tu_A - k - l \ge 2T - k - l > lb, \qquad lt_B = k - Tv_B \ge k + 2T > lb.$$

Both odd candidates follow the collision. Also $T - b = k(k-q-2) + l(k+l-6) + 1 > 0$, excluding an earlier even parent point.

In the other regions, $k \equiv \pm w \pmod{l}$, $l = sw + d$. For $1 \le j \le s$, $w \le wj \le l - d \le l - 2$; for $j = s+1$, $wj = l + (w - d)$ with $2 \le w - d \le w - 2 < l - 1$. Thus none of $1, \dots, s+1$ gives $wj \equiv \pm 1 \pmod{l}$. Equation (5) yields $u_A, -v_B \ge s + 2$. The last bound of (8), or the fourth of (10), again gives $h, t_B > b$, while $T - b > 0$ excludes even parent points. Apply this jointly with continuation of the development until a cone: there is no other fine lattice point inside a unit triangle, and no cone inside a parent edge. No hidden cone crossing is assumed away. Both representative starts are excluded in every negative branch; the two-orbit reduction excludes all 120 starts. Combining the positive constructions, the region table, and the nonprimitive obstruction proves the stated classification, subject to the review status in Section 11.

8. Three consequences

8.1. **The smallest primitive negative example.**

Corollary 2. *Among primitive parameters $k > l > 0$, the smallest negative A3 example is $(17, 5)$, with $T = 399$ and $20T = 7980$.*

Machine-readable claim: `cor:minimum`.

This is not minimality among all fullerenes or among nonprimitive A3 images. All 70 primitive pairs with $T < 399$ satisfy (2); $(17, 5)$ has quotient three and remainder two. Of the 123 pairs with $T \le 673$, the seven negatives are

$$(17, 5), (18, 5), (19, 7), (22, 5), (17, 12), (23, 5), (21, 8).$$

The bound reduces enumeration to $k \le 25$, $0 < l < k$. Claude checked all 123 on an independent construction (T0669); the other 116 were positive. At C_{7980} the two orbit repeats were $228 \to 15$ and $229 \to 14$. See `A3-minimum-AP.md` and Section 10.3.

8.2. **Density of primitive parameter points.**

Corollary 3. *Let $D(X)$ count primitive pairs $k > l > 0$ with $T \le X$, and let $P(X), Z(X)$ count the positive and negative pairs. Then*

$$\begin{aligned} D(X) &= \frac{X}{\pi\sqrt{3}} + O(\sqrt{X}\log X), \\ P(X) &= \Theta(\sqrt{X}\log X), \\ \frac{Z(X)}{D(X)} &= 1 - \Theta(\log X/\sqrt{X}) \longrightarrow 1. \end{aligned} \tag{12}$$

Machine-readable claim: `cor:density`.

These are parameter counts, not a uniform measure on all fullerenes or on graph isomorphism classes. The general counting argument remains a proof draft, without complete independent review or formalization. By the owner's decision of 19 October 2026, this density corollary is excluded from the formalization target. Its counting, Möbius-inversion, error-bound, and limit machinery forms a separate substantial project; the current priority is the graph classification, minimum, and affine-ray corollaries. The prose proof and its unresolved review status are retained. Put $K = \lfloor\sqrt{X}\rfloor$. Separate $m = 1$, $k = qm \pm 1$, and the second-stage cases $r = 1$ or $m = sr \pm 1$, $k = (2s+1)r \pm 2$. There are at most two choices of l for each (k, m), giving, with $H_K = \sum_{j=1}^{K} 1/j$,

$$P(X) \le 4K + (6K+8)H_K = O(\sqrt{X}\log X).$$

For a lower bound, let $L = \lfloor\sqrt{X/3}\rfloor$ and $M = \lfloor(L-1)/3\rfloor$. Count $(k, l) = (qm+1, m)$ with $2 \le m \le M$ and $3 \le q \le \lfloor(L-1)/m\rfloor$. These points are primitive and positive and satisfy $k \le L$, $T \le 3L^2 \le X$. Their number is $(L-1)(H_M - 1) + O(M) = \Omega(\sqrt{X}\log X)$. The region $k > l > 0$, $T \le X$ has area $\pi X/(6\sqrt{3})$, with lattice-point error $O(\sqrt{X}+1)$. Möbius inversion, $\sum \mu(d)/d^2 = 6/\pi^2$, gives the formula for $D(X)$ with error $O(\sqrt{X}\log X)$, and hence (12).

8.3. **Eventual periodicity on integer affine rays.**

Corollary 4. *Let $k(t) = at + b$, $l(t) = ct + d$ have integer coefficients and satisfy $k(t) > l(t) > 0$ for all sufficiently large integer t. The existence sequence is eventually periodic, with an arithmetically computable threshold and period.*

Machine-readable claim: `cor:rays`.

This is a consequence of Theorem 1, not a separate novelty claim. If $\Delta = ad - bc \neq 0$, then $\gcd(k, l) = \gcd(k, l, \Delta)$, so primitivity depends only on $t \bmod |\Delta|$. If $\Delta = 0$, a nonconstant ray is an integer multiple of a fixed primitive direction, with multiplier eventually at least two, and is negative. A constant ray is constant. Eventually $m = \min(l, k - l)$ is one affine expression. Constant m gives fixed-modulus congruences. If m grows, $\lfloor k/m\rfloor$ stabilizes; even when the limiting ratio is an integer, the constant sign of the remainder selects the eventual quotient. At quotient two, $r = k - 2m$ is affine; persistent $r = 0$ is nonprimitive and $r = 1$ is positive. The remaining conditions are affine divisibility tests.

For $A(t) = \alpha t + \beta > 0$, $B(t) = \gamma t + \delta$ with $\alpha > 0$, $A \mid B$ implies $A \mid (\alpha\delta - \gamma\beta)$. If that constant is nonzero, divisibility eventually fails. If it is zero, $B = (\gamma/\alpha)A$ and divisibility holds identically exactly when γ/α is an integer. For $\alpha = 0$ it is a fixed-modulus congruence. Combine the inequality thresholds and take the least common multiple of the fixed moduli and $|\Delta|$. This is the direct argument in `A3-ray-theorem-AU.md`; after classification, the conclusion also falls under parametric Presburger arithmetic [9].

9. Correspondence with Lean (P001/P002)

9.1. **Exact graph specification and completed conclusions.**

The fixed environment is Lean 4.35.0 and mathlib v4.35.0. The coordinator wrote an independent specification using only mathlib: twenty oriented parent faces on twelve labels, integer lattice points of closed triangles, and

equivalence generated by edge gluings. The simple graph `ClaudeChallenge.gcGraph` defines adjacency from lattice triangles, and `LiteralPath` uses actual neighbour sets, distinct vertices and an even number of vertices. Internal degree-five vertices are allowed. The arithmetic predicate `ClaudeChallenge.criterion` includes $\gcd(k,l) = 1$ and the alternatives in (2). Domain assumptions are $k, l \in \mathbb{N}$, $0 < l < k$; positive integer inputs correspond to the paper's domain.

The research team's surface descriptions are connected by proved graph isomorphisms to this independent quotient. In the nonprimitive case the closed model retains ordinary lattice vertices on parent edges. Degree five is proved equivalent to being a parent vertex, and other vertices have degree six. The argument derives local triangle continuation, coherent chart changes and a finite lift of every literal path. It is not conditional on an unproved development certificate. Primitive necessity uses the start reduction and sector collisions; the coordinator's positive construction supplies the converse. The resulting theorem is `ClaudeMain.main_full` in `formal/Formal/ClaudeMainFull.lean`:

$$(\exists n, p,\ \mathrm{LiteralPath}(\mathrm{gcGraph}(k,l), n, p)) \quad \Longleftrightarrow \quad \mathrm{criterion}(k,l).$$

There is no primitivity hypothesis in this theorem. The declaration `ClaudeMain.main_primitive` is its earlier primitive checkpoint. The full declaration proves the independent challenge's statement `ClaudeChallenge.main`; it does not add an axiom with that name.

For Corollary 2, `ClaudeCor.minimum_at` proves that among primitive $k > l > 0$ with $T \le 399$, absence is equivalent to $(k,l) = (17,5)$. `minimum_negative`, `minimum_below` and `minimum_data` record absence, existence below the bound, primitivity and the norm calculation. The bounded arithmetic table uses kernel `decide`, not native evaluation.

For Corollary 4, `ClaudeCor.criterion_ray` proves existence of a threshold and a positive period on every integer affine ray eventually satisfying $0 < l(t) < k(t)$. `R002.A3Release.path_ray` transports this to literal-path existence using the full classification, including nonprimitive points. The parameter t is natural, which describes the same positive tail as sufficiently large integer t. No executable algorithm computing the threshold and period is certified. Therefore the entire wording of Corollary 4 retains a partial formalization obligation. Corollary 3 remains outside the formalization target. No formal proof of the Python or C solver, the census generators, or the order formula is inferred.

9.2. **Axioms, reproducibility and Comparator scope.** The packet includes the Lean source dependency closure, the pinned toolchain and manifest, and `Formal/A3ReleaseCheck.lean`. The latter reprints axiom dependencies and runs an automatic allowlist gate accepting only `propext`, `Classical.choice` and `Quot.sound` transitively. It rejects unfinished proofs, custom axioms and native-evaluation trust axioms. Reproduction instructions and exact records are in `FORMALIZATION.md` and `formalization.json`; mathlib caches are excluded. The four small Python packet checks inspect saved types and outputs; they are not a Lean execution.

The coordinator's Comparator challenge copies its independently written definitions and has a deliberately unproved target statement. The submitted solution is `ClaudeMain.main_full`. Comparator checks the equality of the challenge statement and the proved statement, with the three-axiom allowlist. The full-theorem run recorded acceptance by Lean's standard and paranoid kernels, lean4lean, nanoda, con-leche and con-ron. The six kernels check the same formal proof; they are not six independent mathematical reviews. That run covers the full classification, including nonprimitive parameters, but not the later minimum and ray proofs. The earlier primitive Comparator run has its separate scope and logs. The macOS runs explicitly disabled the unavailable Linux sandbox; they do not establish process isolation. The fixed reports and raw outputs are supplied, without claiming a new six-kernel run in v7.

An independent specification and a proved graph isomorphism reduce the risk of proving a convenient surrogate. They do not prove that the specification expresses the intended published definition or that every coordinate formula in the prose is a transcription of the formal proof.

On 19 October 2026, the owner reported checking the independent challenge statement with the Japanese reading guide and confirming that it expresses the paper's classification claim (P006-15). This is a human check of the statement's meaning, not human peer review of the proof, a new machine equivalence proof, or a novelty review. Human proof peer review and novelty relative to the full position-mapping theory remain outstanding. The historical 41 arithmetic lemmas and four conditional/prototype entries are retained with their exact types in the 57-claim inventory. Their count is not a percentage of formalization; they are no longer the frontier of the supplied formal development.

10. Finite checks and selection history

10.1. **Predictions and comparisons.** Discovery data and later fixed predictions are distinguished; successful later tests do not erase failed conjectures or selection history.

Stage	Scope and result
AP/T0669	All 123 primitive pairs through $T = 673$, rechecked after discovery on an independent construction: 116 positive, seven negative. These are not 123 unseen tests.
AR	Twenty points fixed at `23486766`: 12 positive, eight negative, all matching.
AS	The 921 points through $T = 5000$ were used for discovery (602 positive, 319 negative). Fourteen separate predictions at `e379785c` all matched. The three counterexamples to AQ's stronger threshold remain in the history.
Claude T0708	All 360 pairs with $5000 < T \leq 7000$, independently constructed: 171 positive, 189 negative. Twenty-six were already known, so 334 were new.
AT	Forty-six points at `15e419a4`, $9049 \leq T \leq 185851$: 24 positive, 22 negative, all matching.
AU	Thirty points at `1ede36c9`, maximum $T = 488721$: 12 positive, 18 negative; all decisions, 12 endpoints, and 36 negative-start collisions matched.
AV	Cutoffs $10^4, 10^5, 10^6, 10^7$ and the checking method fixed at `bba3e303`, without predictions of exact finite counts. Two arithmetic implementations agreed; no new graph census was run.

At $X = 10^7$, the arithmetic count was 1,837,762 primitive points, 81,946 positive and 1,755,816 negative (about 95.54%). This is not a proof of the limit or of monotonicity. Direct endpoint formulas were compared with unit continuation on 161 AR, 76 AT, and 323 AU paths. These 560 paths include repeated graphs.

10.2. **Belt tables and the AW package.** Claude independently recalculated the eight belt tables in T0743. Execution of `review-AW` is recorded in AX: T0746 checked eight belts and 20 caps; T0747 checked 337 endpoint orders (159 outer, 63 c2, 68 c1, 47 folded second); T0749–T0768 checked every vertex, cut triangle piece, and cut for 160 known negative graphs and 320 starts. T0773–T0782 checked all 19,200 starts of those same graphs. T0770 compared 1809 arithmetic values on nine rays; T0771 checked 14 graph instances. Probes T0748/T0769 are not additional examples. T0772 stopped before execution on a resource-limit error. The 37 trials comprise 36 successes and one technical failure: 178.167 seconds total, maximum 13.817 seconds per trial and 97.3 MB, no charge. The aggregate 286,940 vertices, 286,300 triangles, 303,857 pieces, and 19,412 cuts are workload counts, not independent examples. Codex wrote the checking code with knowledge of the proof; Claude ran it on an independent constructor. This is finite verification, not an independent general proof.

10.3. **Reports received from R004.** Earlier general integer-GC results use the completed reports 3–6 at `234253238be3949656fa44d63829bbc3e16eb4a3`, fixed in AM. Reports 9–10 were received at `da213c7c`, with byte-identical copies in `A3-v3-received`. For the census, report 9 established exact agreement through C_{176}, with 19,703,439 outputs per generator and 228 negatives. R004 compared complete canonical-code sets, including positive graphs. Its Colab checks reread every stored graph and its decision evidence; Mac checks covered hashes, decision rows, full code sets, and negative certificates, not a fresh check of every positive path on the Mac. Report 10's partial C_{178} recovery (34 of 256 partitions, 940,264 graphs, no negatives) was not a completed census.

Independent census receipt. R004 report 11 confirms complete canonical-code-set agreement between its two generators for all 64 even sizes C_{60}–C_{186}: 43,636,843 outputs per generator and 346 negatives. The coordinator compared its size table with the main team's fixed table, size by size, and found complete count agreement; the fixed receipt is `coordinator-C186-check.original.log`. This edit independently recounted those stored tables, without regenerating graphs or comparing the code streams again. The range C_{188}–C_{200} is outside R004's authorized comparison scope and remains a main-team-only result. The main team's full totals of 120,622,224 outputs per generator and 816 negatives are unchanged. This is a dated receipt boundary, not a claim about live jobs.

R004 report 10 checked the A3 claim using a GC construction implemented from the original source, both adjacency search and face-strip tracking. For $T \le 700$, the theorem domain $k > l > 0$ contains 191 checked pairs: 127 primitive (120 positive, seven negative) and 64 nonprimitive (all negative). A further 39 nonprimitive boundary pairs, 25 of the form $(k, 0)$ and 14 of the form (k, k), are outside that domain and also negative. Thus there are 230 non-control cases in total. The two controls $(1, 0), (1, 1)$ give 232 stored records. The coordinator's v5 arithmetic review corrected the earlier description of all 230 as theorem-domain cases; see correction P006-09. All decisions matched, including minimality at $T = 399$. The report reread all records, positive paths, and 13,200 negative-start certificates. The main team's proof and code were not read, but the expected claim was known: this was not blind. The same verifier used multiple implementations; it was not a further agent's audit. Editing this paper did not rerun those constructions.

R004 independently reduced the nonprimitive case to integer GC inflation and proved the positive belt branch $k \equiv \pm 1 \pmod{l}$ (`df84a3d2`, corrected coordinator receipt `0c59d19f`). The newly received reports 12 and 13 extend this to all negative and all remaining positive cases, respectively. Their proof map also derives the primitive minimum $(17, 5)$, $T = 399$, by general bounds. These fixed receipts are `R004-report-012.md`, `R004-report-013.md`, and `R004-a3-claim13-proof-map.md`. The verifier knew the claim but did not consult the main team's proof, construction or code: this is independent derivation, not a blind test. The coordinator's two fixed reviews report no gap; they are reviews by one AI, not human peer review or formalization. P006-13 records this change of receipt state. The census receipt boundary above is separate.

10.4. **Route of the independent general proof.** The negative proof reduces the 120 starts to two orbits of 60 under icosahedral rotations. Its developed rows are, for integer t,

$$p^+_{2t} = (t, 0), \quad p^+_{2t+1} = (t, 1), \qquad p^-_{2t} = (t, 0), \quad p^-_{2t+1} = (t + 1, -1).$$

After the belt region, the other sectors use four- or six-face blocks. Face-label permutations of order three or two identify a twelve-face period. The proof supplies face containment, gluing and exclusion of intermediate parent vertices. With the sector-dependent integers A, J defined in `R004-a3-remaining-necessity-proof.md`, the return maps are written here as $\mathcal{G}, \mathcal{H}$ to distinguish them from our GC coordinate map:

$$\mathcal{G}(x, y) = (A - y, x + y - J), \qquad \mathcal{H}(x, y) = (x + y + A - J, -x + J).$$

They give the following explicit equalities of surface vertices:

	upper row	lower row
$\mathcal{G}$	$2J-1 \mapsto 2A-2$	$2J-2 \mapsto 2A-1$
$\mathcal{H}$	$2J-2 \mapsto 2A-1$	$2J-1 \mapsto 2A-2$

The sector inequalities put a repeated vertex before any legal endpoint; the argument needs such a repeat, not a proof that it is the first repeat. Together with the belt and nonprimitive arguments this excludes all negative parameters, including paths that might otherwise meet an internal degree-five vertex.

For the remaining positive sectors, `R004-a3-positive-remaining-proof.md` constructs endpoints $E = (A, \sigma)$, $\sigma = \pm 1$, in a strip of width one. Four-face blocks with an order-three permutation, or six-face blocks with an order-two permutation, run through twelve distinct parent faces. Copies of a face are separated transversely by $-3d, -2d$, or $2d$ with $d \geq 2$. The proof checks the initial and terminal caps as well; primitive parent edges have no interior lattice points. These facts give injectivity, distinct endpoint labels, actual endpoint degrees, and both consecutive and skip adjacency. The two row constructions have $2A + 2$ or $2A$ vertices, respectively. Small widths and boundary ratios are handled by the previously proved belt branch.

This route has separately derived face tables and repeat formulas; it does not independently certify every coordinate convention of our Appendices B and C. Both arguments use developed strips, belts and transverse width, so mathematical nonequivalence is not asserted. The fixed coordinator reviews describe symbolic checks and finite spot checks, with some containment and cap checks relying on the prose argument and sampled instances. They also retain the external nonfacial-three-cycle prerequisite. Editing this summary did not rerun those checks or add another independent reviewer. The reviews are `R004-coordinator-review-a3-necessity-2026-10-19.md` and `R004-coordinator-review-a3-positive-2026-10-19.md`.

10.5. **Additional checks and the $T = 1099$ separation.** The `review-A3` package was fixed before execution at `cb01852c`. Claude ran Codex's checks on its independent graphs. T0789 verified 24 gluing records (10 outer, 10 positive-side, four negative-side), 82 edge moves, 48 terminal triangles, and 144 vertex images. T0795–T0864 checked 560 known positive paths on 417 distinct graphs, including 584,420 vertices and 583,300 unit triangles. T0865–T0884 checked 159 known outer-negative graphs, 318 starts, including repeats, prior-cone exclusion, regions, cuts, and edge gluings. The five probes T0790–T0794 overlap these samples. T0789–T0884 contains *96*, not 95, successes: one atlas, five probes, and 90 batches. This corrected addition is recorded in `A3-v2-validation.json`. Total time was 234.914 seconds (231.601 for the batches), maximum 7.034 seconds per trial and 103.8 MB, no charge. T0788 was a pre-execution argument-expansion failure. None of these known examples is counted again as an unseen test.

R005 compared $(23, 15)$ and $(27, 10)$: both have $T = 1099$, the same final group product u on 60 states, and six closed zigzags of length 10990. Formula (2) nevertheless labels the first positive and the second negative. This comparison initially used arithmetic, not graph decisions. Claude subsequently constructed both C_{21980} graphs (10,992 dual vertices, IPR) independently, finding a 290-vertex path in the first and no path in the second. It did not recalculate the zigzag invariants; this rewrite did not rerun the graph tests. The pair was selected during literature comparison, not preregistered as an unseen sample. It shows that existence is not a function of (T, u) alone. It does not exclude a derivation using the full position mapping, z-vectors, or intersection order. R005's later fixed report also finds that the type-I/type-II self-intersection signatures differ: $(320, 590)$ versus $(270, 640)$. Thus this example separates only the compared counts, lengths and pentagon-contact data, not the full intersection theory. This revision receives that finite check without rerunning it.

11. Review obligations and related work

11.1. **Current status of the claims.** The definition, two start orbits, and candidate endpoints have been checked against the source and in AI prose review, but lack a complete formal surface connection at that historical checkpoint. The main AR positive argument was reviewed; its finite belt and cap tables were checked later. Through the fixed review `9fedf3fc`, the coordinator checked congruence exclusion, endpoint ordering, the positive-side A calculations, outer A coordinates, negative-side A regions, the collision identities, and their period differences. Ownership of all outer A pieces and some negative-side A lower bounds retained abbreviated steps. The periodic transfer of the finite tables was rederived. Signs, small widths, simplicity, and degrees were checked, while some full cap-piece arguments remained partial. Appendices B and C explicitly supply B's three regions and the terminal caps. A later fixed review, `49430e1c` (03:46 JST, 19 October 2026), records the coordinator's recalculation of all formulas in the Japanese v4 Appendices B and C, with no missing steps found. This review was omitted from English v6's status metadata. English v6.2 corrects the omission; it transcribes the Japanese proof, with the coordinator's separate machine checks of selected English formulas recorded in P006. The later addendum `c1b27f1d`, headed 15:10 JST, completes the general AI prose review of A-sheet items 2 and 3: Appendix C supplies the terminal containment, and the diagonal-A initial cap was rederived in that addendum. P006-11 records this update. The English-v5 review addendum `f88d1b6a` identifies its one failed check as the R004 nonprimitive domain count corrected in P006-09; the other 76 checks passed. The typesetting issue was outside those 77 checks. Neither addendum supplies human peer review, full formalization, or an independent necessity proof; the later R004 receipt below supplies the last of these separately.

P006-13 records two independently derived general prose proofs of the entire classification: the research team's proof and R004's alternative. R004 knew the claim but did not read the main proof, construction or code. One AI coordinator reviewed both directions; this is not human peer review. P006-14 adds the full Lean theorem and the primitive minimum, and records the full-theorem Comparator acceptance. Thus `thm:classification` and `cor:minimum` are `lean-proved` for the exact independent graph statements, while novelty and human proof review remain separate obligations. P006-15 records the owner's statement-meaning confirmation without promoting the proof to human-reviewed. Ray periodicity has a formal existence theorem; effective threshold and period computation remains uncertified. Density retains its general counting review obligations. These updates do not extend the separate dated census receipt boundary in Section 10.3.

11.2. **Related work and the novelty boundary.** The problem comes from the classification question for fullerenes admitting no generalized Stone–Wales operation in [1]. Our literal Definition 1(a) scope and the restriction to interior A3 parameters are essential. The Goldberg–Coxeter construction, tightness of primitive images, and arithmetic of closed zigzags are established theory. Dutour–Deza's position mapping uses a directed parent-dual edge and a position p; its group product determines zigzag number and length and can be computed by a Euclidean algorithm [2, Section 4, Theorems 4.5, 4.12, 6.5 and Proposition 4.14]. For primitive dodecahedral images, the normalized partition vector has three possible types. Zigzags, railroads, and their self- and mutual intersections have also been studied [3, 8]. None of these constructions or methods is itself claimed as new here.

The compared counts, lengths and pentagon-contact sequences alone do not determine the existence of a simple open gSW path. In R005's comparison of 489 already known primitive pairs, adding the parent-distance sequence of pentagon contacts or the counts of shared pentagons still did not separate the two-stage decisions. At $T = 1099$, even the complete pentagon-contact cycles of all zigzags, in the fixed parent labels and up to cyclic start and reversal, agreed for the pair above. However, R005's later check found different self-intersection signatures with types I and II kept separate: $(320, 590)$ and $(270, 640)$, respectively. This is not a counterexample

to determination from the complete z-vector or full intersection order. P006-16 and the supplied fixed report `R005-novelty-A3-deep-2026-10-19.md.original.log` record the correction. Its two tracing implementations agreed on five known examples, sharing the GC constructor; that finite check is not an independent construction or a general classification proof. The 489 labels were obtained from the AU formula, not a new census of graphs. The report is `R005/report-2026-10-19-position-mapping.md`.

Full timing information can recover the input. Put $n = k+l$ and let τ be the first position-update count from $p = 1$ to $p = n$. Then $\tau l \equiv -1 \pmod{n}$, so the unordered adjacent intervals $\{\tau, n-\tau\}$ determine (k, l) under $k > l > 0$. Recovering the input alone is not a short deduction of gSW existence. The additional geometric task is to preserve legal endpoints, parity, and the order of revisits to original dual vertices, and reduce their stopping condition to the two residue stages.

Comparing self-intersection against arrival at the next pentagon also has precedent: [4, Section 6, Fig. 5] uses such a comparison for Coxeter coordinates $(7, 2)$, C_{1340}, to exclude a particular twisted Endo–Kroto reduction patch. That path and operation differ from literal gSW. Catalogues of isomerization patches [5], Caspar–Klug T- and gcd-based classifications [6], and congruence conditions in a pentagonal-polyhedron model [7] are related, but the portions examined did not supply this same gSW criterion. Reading of [5] was limited to its abstract; [7] was read mainly in its abstract and Section 2. The stored literature review has now examined the full twelve-page body of Deza–Dutour–Fowler (2004), *Zigzags, Railroads, and Knots in Fullerenes* [8], pp. 1282–1293. In the text examined, it did not locate this same two-stage literal-gSW classification. Its GC/gcd, zigzag and tightness results do not by themselves supply the endpoint-before-revisit comparison used here; a short derivation from the full intersection theory has not been excluded. Sections 2.5–2.6 and 7.1–7.6 of *Geometric Structure of Chemistry-Relevant Graphs* (2015) remain unread; contents and chapter abstracts are not full-text review. Novelty is not established.

The candidate contribution is the reduction, over all interior A3 parameters, of the comparison between the first legal pentagonal endpoint and the first revisit to the same original face to an explicit two-stage residue criterion. More precisely, from a legal start let R be the first repeated original face index and E the first odd index at least three with a pentagonal face preceded by a hexagonal face, before imposing simplicity ($E = \infty$ if none exists). A literal path exists from that start exactly when $E < R$. Internal pentagons are allowed. Known flag or position-mapping encodings already permit a finite search; the candidate difference is the closed arithmetic condition for some legal start, with geometric positive witnesses and negative revisits. Neither GC construction, gcd decomposition, Euclidean group products nor finite decidability is proposed as a new principle. The search through 19 October 2026 did not locate the same criterion within the material actually examined. Whether it is a short consequence of the full position-mapping and intersection theory remains open; no priority or worldwide novelty guarantee is made. After classification, ray periodicity is a consequence of parametric Presburger arithmetic [9, Theorem 1.15, Property 1], not a new general periodicity principle.

Appendix A. IPR census from C_{60} through C_{200}

Machine-readable claim: `app:census`.

A.1. **Scope, method, and preservation.** IPR means that no two pentagons share an edge. The 71 even sizes $N = 60, 62, \ldots, 200$ were enumerated with buckygen 1.1 and fullgen from plantri 5.8. The saved cumulative record P contains 120,622,224 outputs per generator. The pipeline checked the dual, spherical rotation system, degrees, IPR, and the classifier's local prerequisites. Negative canonical-code sets agreed at every size, giving 816 distinct classes. The main team did *not* compare full canonical-code sets of all positive graphs. Using two generators does not independently validate their shared decision code.

Before adopting the C classifier, the C–Python audit compared it with both forced continuation and a definition-based DFS on 1,456,598 general C_{20}–C_{100} outputs and 4,770,156 IPR C_{60}–C_{160} outputs: 6,226,754 in total, all agreeing, with positive witnesses checked against the definition. These were buckygen outputs and the two ranges overlap. This is not a Python rerun of all 120,622,224 outputs. Subsequent negatives were checked by both Python searches and exhaustive-start absence certificates. Generator completeness and correctness on every possible input are not formally proved by these tests.

Saved records include size/generator/partition counts and completion, implementation hashes, negative full codes and adjacency lists, 16 audits and cumulative summaries, and one JSON certificate per negative class. A certificate contains the dual adjacency and the maximal simple forced path and stopping reason for every admissible initial triangle. The index's `canonical_sha256` hashes canonical-code bytes; `certificate_sha256` hashes the JSON file. The original audits compared the full codes, not just their hashes. The v3 metadata extraction verified saved hashes and counts without rerunning search, canonicalization, or certificate logic.

The entire census was not one preregistered experiment: it includes discovery and later confirmation. Predictions of the full negative sets for C_{186}–C_{200} were fixed at `3ac5c0b` and matched unchanged. Failures, interruptions, and selections are preserved in report P and its trials.

A.2. **Counts by size.** The following table transcribes the frozen `A3-v3-data/counts.csv`. Zero total output and zero negatives among existing graphs are distinct.

Carbon order	Outputs per generator	Negative classes
60	1	0
62	0	0
64	0	0
66	0	0
68	0	0
70	1	0
72	1	0
74	1	0
76	2	0
78	5	0
80	7	1
82	9	0
84	24	0
86	19	0
88	35	0
90	46	0
92	86	0
94	134	0
96	187	1
98	259	0
100	450	0
102	616	0
104	823	1
106	1,233	0
108	1,799	0
110	2,355	0
112	3,342	2
114	4,468	0

Carbon order	Outputs per generator	Negative classes
116	6,063	0
118	8,148	0
120	10,774	3
122	13,977	0
124	18,769	0
126	23,589	0
128	30,683	6
130	39,393	0
132	49,878	0
134	62,372	0
136	79,362	6
138	98,541	0
140	121,354	0
142	151,201	0
144	186,611	15
146	225,245	0
148	277,930	0
150	335,569	0
152	404,667	17
154	489,646	0
156	586,264	0
158	697,720	0
160	836,497	40
162	989,495	0
164	1,170,157	1
166	1,382,953	0
168	1,628,029	45
170	1,902,265	1
172	2,234,133	0
174	2,601,868	0
176	3,024,383	89
178	3,516,365	0
180	4,071,832	1
182	4,690,880	0
184	5,424,777	117
186	6,229,550	0
188	7,144,091	0
190	8,187,581	0
192	9,364,975	199
194	10,659,863	0
196	12,163,298	0
198	13,809,901	0
200	15,655,672	271
Total	120,622,224	816

A.3. **The 816 classes and their relation to A3.** The saved taxonomy U assigns disjoint explanations:

Explanation	Classes
GC$(2,0)$ images	812
GC$(3,0)[C_{20}]$, at C_{180}	1
$C1(2,3) = F(2,3)$, at C_{164}	1
$O2(2,3) = G(2,2,3)$, at C_{170}	1
$C1(3,2) = F(3,2)$, at C_{184}	1
Unexplained in this finite census	0

The GC2 comparison includes 116 at C_{184}, 199 at C_{192}, and 271 at C_{200}. The three noninteger-GC examples were identified by full canonical-code equality with their constructions. Here "noninteger GC" means not GC$(d,0)[F]$ for an ordinary fullerene F and $d \geq 2$, not the complement of all GC images. The C1 and O2 identifications are with existing catalogue families, not new family constructions. Their general absence arguments are separate from the individual certificates and this finite accounting.

In the theorem's interior domain, $20T \leq 200$ permits only $(2,1)$, $T = 7$, C_{140}, which is IPR and positive, consistently with $m = 1$. All 121,354 IPR C_{140} outputs are positive. The boundary points $C_{60} = \mathrm{GC}(1,1)[C_{20}]$, $C_{80} = \mathrm{GC}(2,0)[C_{20}]$, and $C_{180} = \mathrm{GC}(3,0)[C_{20}]$ are not interior-theorem cases. The census therefore does not verify a broad range of interior A3 parameters.

A.4. **Independent comparison and the local data package.** Section 10.3 fixes the independent receipt boundary at C_{186}: 64 even sizes, 43,636,843 outputs per generator and 346 negatives, with the coordinator's size-by-size count comparison received. The range C_{188}–C_{200} is a main-team-only result. The published-data plan adopted for local preparation contains an English README and schema, a 71-row CSV, an 816-row JSONL index, 816 certificate JSON files, audit summaries, decision and checking code with dependency hashes, and generator metadata. The certificate files total 60,758,514 uncompressed bytes. Generator source and binaries, raw output streams, and all positive graphs are excluded. Package-relative references retain both original and derived hashes. Local archive hashes and the exact validation scope are recorded separately; no archive is uploaded by this preparation. Authorship and licenses are set below. Third-party terms are retained; journal and arXiv submission are separate later decisions.

Appendix B. The negative B orientation: coordinates, cuts, and ownership

Machine-readable claim: `app:B`.

This appendix expands Section 6 without new parameter conditions. It covers every piece of every fine triangle, not just vertices. Assign piece interiors to parent faces and use the agreement of neighbouring affine gluing maps on boundaries. The facial-triangle prerequisite, primitive parent edges, and the exclusion of earlier parent vertices are as in the body and Section 7.

B.1. **Two coordinate systems.** Let (x, y) be B's original fine coordinates and (α, β) its original parent coordinates. Its triangle sequence lies in $0 \le x \le b$, $0 \le y \le 1$, with $T\alpha = (k+l)x + ly$, $T\beta = -lx + ky$. For the two slanted fan types the actual faces are rotated by R^{-1}. For calculation only, rotate back to common coordinates:

$$\begin{gathered}(X, Y) = (-y, x+y), \qquad (u, v) = (-\beta, \alpha+\beta),\\ Tu = lx - ky, \quad Tv = kx + (k+l)y, \quad T(u+v) = (k+l)x + ly.\end{gathered} \tag{B1}$$

Thus $-1 \le X \le 0$, $0 \le X+Y \le b$, not A's rectangle. The original parent-face names are recovered by

$$R^{-1}U(i,j) = D(i+j, -i-1), \quad R^{-1}D(i,j) = U(i+j+1, -i-1). \tag{B2}$$

This planar coordinate change does not rotate quotient labels; use the actual B gluing tables for each face map.

B.2. **Outer fan: $k = ql + r$.** Here retain (α, β), with strip $-1 \le \beta \le 0$. Call the six positive expressions in Section 6.1 $F_1, \dots, F_6$. Throughout the rectangle,

$$\beta \ge -lb/T > -2, \quad \beta \le k/T < 1, \quad \alpha + \beta \le (kb + k + l)/T < q + 2.$$

The first bound uses $2T - lb = F_1 + k + l$ and the last uses F_3. If $\beta > 0$, then $x < ky/l$, $\alpha \ge 0$, and $\alpha + \beta = [kx + (k+l)y]/T < y/l \le 1/l < 1$, placing the initial piece in $U(0,0)$. At $\beta = 0$, $\alpha = y/l$. At $\beta = -1$,

$$\alpha = q + 1 + (r+y)/l \in (q+1, q+2),$$

using $r \ge 2$, $r + 1 \le l - 1$; the cut beyond $(q+2, -1)$ is avoided. If $\beta < -1$, then $\alpha + \beta = (y - k\beta)/l > k/l > q$. The resulting region $-2 < \beta < -1$, $q < \alpha + \beta < q + 2$ is covered by

$$D(q+1, -2),\ U(q+2, -2),\ D(q+2, -2),\ U(q+3, -2).$$

The line $\alpha+\beta = q+1$ splits the first pair from the second; $\alpha = q+2$ or $q+3$ identifies the individual triangle. The first crossing is the common edge of $U(q+1, -1)$ and $D(q+1, -2)$. At $(b, 1)$,

$$T(-1-\beta) = F_6 > 0, \qquad T(\alpha + \beta - q - 1) = F_5 + k + l > 0.$$

Thus H_B applies on the last two faces. Its translation differs from the B translation in (7) by $-(5, 0)$. The early point $(a, 0)$ has $\beta = -la/T$ and $T - la = F_5 + k + l > 0$, so it is inside the old belt. Together with ordering and Section 7, this gives $2b + 1 \to 2a$.

B.3. **Positive-side fans: $k = cl + w$, $l = sw + d$.** Let $c = 1, 2$, with $s \ge 2, 1$, $2 \le d \le w - 2$, and $W_v = cs + c + 1$. The common normal is $n = v - cu = \alpha + (c+1)\beta$, with

$$Tn = wx + [(c+1)k + l]y \ge 0, \qquad -y = wu - ln.$$

There is no initial negative-normal B piece. At $n = 1$, $u = s + (d - y)/w \in (s, s+1)$. For $n > 1$, $u > (l-1)/w > s$, so exit is on the cell-s side of the cut from W. Call the quantities (8) $P_1, \dots, P_9$. Their exact uses are:

Required bound	Positive numerator
$v < W_v + 1$	P_4
$u + v < s + W_v + 2$	$P_5 + (k - l)$
$u < s + 2$	$P_9 + k + l$
$v > W_v$ at $(b, 0)$	$P_6 + l$
$u + v > s + W_v + 1$ at $(b, 0)$	P_7
$n < 1$ at $(a, 1)$	P_8
$0 < a < b < T$	P_1, P_2, P_3

For complete ownership, partition the $n > 1$ portion into three cases. If $u \le s + 1, v \le W_v$, then $s < u \le s + 1$ and $cu + 1 < v \le cs + c + 1$, within the original full cell triangles $U(s, j), D(s, j)$, $j = cs, \dots, cs + c$. Here "cell" includes the full glued triangles, not just their intersection with the slanted strip. If $u \le s + 1, v > W_v$, the bound $v < W_v + 1$ gives $U(s, W_v)$ or $D(s, W_v)$, divided by $u + v = s + W_v + 1$. If $u > s + 1$, then $v > W_v$ and $u + v < s + W_v + 2$, giving $U(s + 1, W_v)$. This covers every piece. The last point belongs to the latter two faces. In B's original coordinates the departure and added faces are

$$U(s + W_v, -s - 1);$$
$$D(s + W_v, -s - 1),\ U(s + W_v + 1, -s - 1),\ D(s + W_v + 1, -s - 2).$$

Conjugating the common map gives $H_B(z) = R^{-1}z + (s+1, W_v)$. The difference from the translation $(s - 5, cs + 5)$ in (9) is $(-6, 4 - c)$: minus one B period, $(6, -3)$ or $(6, -2)$. At $(a, 1)$, $0 < n < 1$, so the early point uses the same belt map. This establishes $2b \to 2a + 1$ without an appeal to analogy with A.

B.4. **Negative-side fan: $k = 2l - w$, $l = sw + d$.** Here $s \ge 2$, $2 \le d \le w - 2$, and

$$n = v - 2u = \alpha + 3\beta, \quad Tn = -wx + (3k + l)y, \quad -y = -wu - ln.$$

At $n = -1$, $u = s + (d + y)/w \in (s, s + 1)$, using $d + 1 \le w - 1$. For $n < -1$, $u > (l - 1)/w > s$, before the cut from $(s + 1, 2s + 1)$. Writing (10) as $M_1, \dots, M_8$, the whole rectangle has $u < s + 2$ from $M_4 + k + l$ and $u + v < 3s + 3$ from $M_5 + (k - l)$. At $(b, 1)$ the lower bounds are M_6 for $u > s + 1$ and $M_7 + l$ for $u + v > 3s + 2$. At $(a, 0)$, $T - wa = M_8 + 2k + 3l > 0$ gives $n > -1$; M_1, M_2, M_3 order the indices.

Since $v = (ku + y)/l$, $u > s$ implies $v > 2s - 1 + d/l > 2s - 1$. For $u > s + 1$,

$$v > k(s + 1)/l = 2s + [(s - 1)w + 2d]/l > 2s.$$

If $n < -1, u \le s+1$, then $2s-1 < v < 2u-1 \le 2s+1$, within $U(s, j), D(s, j)$ for $j = 2s-1, \dots, 2s+1$. If $u > s+1$, the bounds put the point in $U(s+1, 2s), D(s+1, 2s), U(s+1, 2s+1)$: below $v = 2s+1$, split at $u + v = 3s + 2$; above it, use the last U. The last point is in the final two faces. Under (B2), these faces are

$$D(s, 2s) \mapsto U(3s + 1, -s - 1),$$
$$U(s + 1, 2s) \mapsto D(3s + 1, -s - 2),$$
$$D(s + 1, 2s) \mapsto U(3s + 2, -s - 2),$$
$$U(s + 1, 2s + 1) \mapsto D(3s + 2, -s - 2).$$

For the initial $n > 0$ piece, return to original coordinates. From $y = ln - w\beta$,

$$0 \le \alpha = \frac{3y - (3l - w)n}{w} < 3/w \le 3/4, \qquad 0 \le \alpha + \beta = \frac{2y - kn}{w} < 2/w \le 1/2.$$

It lies in $U(0, 0) \cup D(0, -1)$, with D central and U additional. Assign $-1 \le n \le 0$ to the central belt. This covers the rectangle. The original map is $H_B(z) = Rz + (2s + 1, -3s - 2)$. The translation

in (11) differs by $(-6,2) = -R^{-1}(2,4)$. The early point has $-1 < n = -wa/T < 0$, proving $2b+1 \to 2a$ at the same ordinary vertex.

Appendix C. Terminal positive caps by point reflection

Machine-readable claim: `app:C`.

C.1. **Action on the sequence and parent triangles.** For $W = (A, B)$ let $J_W(z) = W - z$. In fine coordinates this is $(1, Q) - (x, y)$ for A, $(Q, 1) - (x, y)$ for B, and $(Q-1, 1) - (x, y)$ for AR B. It preserves the relevant rectangle and gives $P_{E-j} = P_E - P_j$. Consecutive triangles and their cut pieces map to the terminal triangles in reverse order. Direct substitution gives

$$J_W U(i,j) = D(A-i-1, B-j-1), \quad J_W D(i,j) = U(A-i-1, B-j-1). \tag{C1}$$

This is planar geometry, not quotient reflection symmetry. Terminal face maps and labels are taken from the fixed cap table.

C.2. **AR B, $k = ql - 1$.** Here $W = (q+1, -1)$, $n = v$, and $n(J_W z) = -1 - n(z)$. The initial $n > 0$ piece has $u \geq 0$, $v > 0$, $u + v \leq y/l \leq 1/l < 1$, so lies in $U(0,0)$. For the terminal $n < -1$ piece, $z' = J_W z$ satisfies

$$u' = q + 1 - u \geq 0, \quad v' = -1 - v > 0, \quad u' + v' = q - u - v \leq 1/l < 1.$$

Thus the terminal cap is $D(q, -2)$, while the initial central $D(0, -1)$ maps to central $U(q, -1)$. Their shared edge is $[(q, -1), (q+1, -1)]$. Reflection preserves the central normal interval. The initial U phase zero and terminal D phase $q - 1 \bmod 5$ have labels distinct from the central belt and each other. Width one was handled earlier; choose the plus branch at width two. Containing each cut piece does not require all vertices of a triangle touching that piece to lie in the same cap face.

C.3. **Second A and diagonal A, $l = sw - 1$.** For $c = 2$, $w = k - 2l$; for $c = 1$, $w = k - l$. Set $n = v - cu$ and $W = (s, cs + 1)$. The minus branch uses $w \geq 3$, and $n(J_W z) = 1 - n(z)$. On the initial $n < 0$ piece,

$$0 \leq u < 1/w, \qquad -u \leq v < cu, \qquad u + v < (c+1)/w \leq 1.$$

It lies in central $U(0,0)$ and additional $D(0, -1)$. Reflecting a terminal $n > 1$ point gives

$$0 \leq s - u < 1/w, \quad -(s-u) \leq cs + 1 - v < c(s-u),$$
$$(c+1)s + 1 - u - v < (c+1)/w \leq 1.$$

The terminal central and additional faces are $D(s-1, cs)$ and $U(s-1, cs+1)$, sharing $[(s-1, cs+1), (s, cs+1)]$. The extra-face phases are $s - 1 \bmod 2$ for second A and $s \bmod 3$ for diagonal A. The fixed table exhausts them. Rectangle invariance and piecewise reflection cover every terminal triangle piece.

C.4. **Folded second B, $l = sw - 1$, $w = 2l - k \geq 3$.** Here $W = (3s - 1, -s)$, $n = u + 3v$, and $n(J_W z) = -1 - n(z)$. For the initial $n > 0$ piece with $0 \leq y \leq 1$,

$$0 \leq u = \frac{3y - (3l - w)n}{w} < 3/w \leq 1, \quad 0 \leq u + v = \frac{2y - kn}{w} < 2/w < 1.$$

Thus $U(0,0)$ is additional and $D(0, -1)$ central. Reflecting the terminal $n < -1$ piece gives

$$0 \leq 3s - 1 - u < 3/w, \quad 0 \leq 2s - 1 - u - v < 2/w, \quad n(z') > 0.$$

The terminal additional face is $D(3s-2, -s-1)$ and the central face $U(3s-2, -s)$, sharing $[(3s-2, -s), (3s-1, -s)]$. The two phases $s \bmod 2$ occur in the table. The central normal interval is preserved and boundary pieces glue along the shared edge.

C.5. **From containment to legality on the surface.** This appendix proves cap containment and gluing as a prose argument. Distinct face labels still depend on the finite table; exclusion of periodic identifications on Section 5.5; and degrees on $Q < T$ and the candidate congruences. It does not by itself complete review of the classification. The Japanese v4 B and C prose received the coordinator's AI review in `49430e1c`; this English transcription and the selected-formula machine checks do not constitute human peer review, a second independent general proof, or full formalization by itself. The separate R004 proof is received in Section 10, rather than established by this transcription.

Main formal declarations

The current entry points are

- `ClaudeMain.main_full`: the full classification, claim `thm:classification`;
- `ClaudeCor.minimum_at`: the primitive minimum, claim `cor:minimum`;
- `ClaudeCor.criterion_ray`: arithmetic eventual periodicity of the criterion on affine rays;
- `R002.A3Release.path_ray`: its transfer to graph path existence, including nonprimitive parameters.

The last two declarations supply the existence clause of `cor:rays`; its certified effective threshold/period algorithm remains outside that formal result. They are not proofs of the density claim `cor:density`.

Historical formal lemma identifiers

These identifiers refer to the exact types and evidence in `A3-v7.3-machine/claims.json`. The following historical list contains the 41 A3 arithmetic declarations, the conditional finite-path theorem, and three selected prototype declarations have distinct entries.

`unit_endpoint`
Machine-readable claim: `lean:unit_endpoint`.
`second_plus`
Machine-readable claim: `lean:second_plus`.
`second_minus`
Machine-readable claim: `lean:second_minus`.
`diagonal_plus`
Machine-readable claim: `lean:diagonal_plus`.
`diagonal_minus`
Machine-readable claim: `lean:diagonal_minus`.
`folded_second_plus`
Machine-readable claim: `lean:folded_second_plus`.
`folded_second_minus`
Machine-readable claim: `lean:folded_second_minus`.
`outer_collisions`
Machine-readable claim: `lean:outer_collisions`.
`positive_fan_collisions`
Machine-readable claim: `lean:positive_fan_collisions`.
`negative_fan_collisions`
Machine-readable claim: `lean:negative_fan_collisions`.
`no_small_multiple`
Machine-readable claim: `lean:no_small_multiple`.
`outer_u_bound`
Machine-readable claim: `lean:outer_u_bound`.
`outer_v_bound`
Machine-readable claim: `lean:outer_v_bound`.

candidate_order
Machine-readable claim: lean:candidate_order.
outer_endpoint_order
Machine-readable claim: lean:outer_endpoint_order.
au_c1_a_pos
Machine-readable claim: lean:au_c1_a_pos.
au_c1_b_minus_a_pos
Machine-readable claim: lean:au_c1_b_minus_a_pos.
au_c1_N_minus_b_pos
Machine-readable claim: lean:au_c1_N_minus_b_pos.
au_c1_v_upper_pos
Machine-readable claim: lean:au_c1_v_upper_pos.
au_c1_sum_upper_pos
Machine-readable claim: lean:au_c1_sum_upper_pos.
au_c1_v_late_lower_pos
Machine-readable claim: lean:au_c1_v_late_lower_pos.
au_c1_sum_late_lower_pos
Machine-readable claim: lean:au_c1_sum_late_lower_pos.
au_c1_early_inside_pos
Machine-readable claim: lean:au_c1_early_inside_pos.
au_c1_endpoint_bound_pos
Machine-readable claim: lean:au_c1_endpoint_bound_pos.
au_c2_a_pos
Machine-readable claim: lean:au_c2_a_pos.
au_c2_b_minus_a_pos
Machine-readable claim: lean:au_c2_b_minus_a_pos.
au_c2_N_minus_b_pos
Machine-readable claim: lean:au_c2_N_minus_b_pos.
au_c2_v_upper_pos
Machine-readable claim: lean:au_c2_v_upper_pos.
au_c2_sum_upper_pos
Machine-readable claim: lean:au_c2_sum_upper_pos.
au_c2_v_late_lower_pos
Machine-readable claim: lean:au_c2_v_late_lower_pos.
au_c2_sum_late_lower_pos
Machine-readable claim: lean:au_c2_sum_late_lower_pos.
au_c2_early_inside_pos
Machine-readable claim: lean:au_c2_early_inside_pos.
au_c2_endpoint_bound_pos
Machine-readable claim: lean:au_c2_endpoint_bound_pos.
au_neg_a_pos
Machine-readable claim: lean:au_neg_a_pos.
au_neg_b_minus_a_pos
Machine-readable claim: lean:au_neg_b_minus_a_pos.
au_neg_N_minus_b_pos
Machine-readable claim: lean:au_neg_N_minus_b_pos.
au_neg_u_upper_pos
Machine-readable claim: lean:au_neg_u_upper_pos.
au_neg_sum_upper_pos
Machine-readable claim: lean:au_neg_sum_upper_pos.

au_neg_u_late_lower_pos
Machine-readable claim: lean:au_neg_u_late_lower_pos.
au_neg_sum_late_lower_pos
Machine-readable claim: lean:au_neg_sum_late_lower_pos.
au_neg_early_inside_pos
Machine-readable claim: lean:au_neg_early_inside_pos.
finite_literal_path_decision
Machine-readable claim: lean:finite_literal_path_decision.
unit_width_path
Machine-readable claim: lean:unit_width_path.
chartAdj_iff
Machine-readable claim: lean:chartAdj_iff.
faces_connected
Machine-readable claim: lean:faces_connected.

Acknowledgements and AI-use statement

The author used Codex and Claude Code to assist with literature organization, mathematical exploration, proof drafting, implementation, finite verification, formalization of the graph classification and specified corollary statements, and English exposition. The author is responsible for the published account, its provenance, and the stated limits of verification. AI systems are not listed as authors. AI prose reviews are identified separately from human review. Only the explicitly listed Lean propositions are described as formally proved; finite checks and independently constructed tests have their stated domains. No claim of complete human checking of every argument is made.

License and contact

Copyright 2026 Tatsuhiko Takimoto. The article text, figures and research data are available under CC BY 4.0; the accompanying code and Lean sources are under Apache-2.0, with third-party notices retained. Contact is through GitHub Issues of the public repository; its URL has one canonical location in the accompanying REPOSITORY-URL.txt. No contact email is listed.

Reproducibility records

Relative repository paths are archival identifiers, not claims of public availability. The earlier proof drafts are A3-unit-residue-proof-AR.md, A3-qge3-proof-AT.md, A3-q2-positive-proof-AT.md, A3-complete-proof-AU.md, and A3-endpoint-order-AU.md. Finite geometry is preserved in review-AW/manifest.json, its cap table, and the three atlases named in Section 6. Review execution hashes are in AX-validation.json, A3-v2-validation.json, A3-v3-validation.json, and A3-v4-validation.json. The English correspondence and validation record accompany this source. The machine-readable claim layer distinguishes dependencies and evidence statuses; its identifiers match the labels above. Hashes establish file identity, not the truth of mathematical contents.

References

[1] A. Bille, V. Buchstaber, and E. Spodarev, *Some Open Mathematical Problems on Fullerenes*, arXiv version 2024; published 2025. doi:10.1021/acs.jcim.4c01997.

[2] M. Dutour and M. Deza, *Goldberg–Coxeter Construction for 3- and 4-valent Plane Graphs*, 2004. doi:10.37236/1773.

[3] M. Deza and M. Dutour, *Zigzag Structures of Simple Two-faced Polyhedra*, arXiv:math/0212352v2, 2003.

[4] G. Brinkmann, D. Franceus, P. W. Fowler, and J. E. Graver, *Growing fullerenes from seed*, 2006. doi:10.1016/j.cplett.2006.07.040.

[5] G. Brinkmann, P. W. Fowler, and C. Justus, *A Catalogue of Isomerization Transformations of Fullerene Polyhedra*, 2003. doi:10.1021/ci020069l.

[6] A. Luque and D. Reguera, *The Structure of Elongated Viral Capsids*, 2010. doi:10.1016/j.bpj.2010.02.051.

[7] F. Sadre-Marandi and P. Das, *Extension of Caspar–Klug theory to higher order pentagonal polyhedra*, 2018. doi:10.1515/cmb-2018-0001.

[8] M. Deza, M. Dutour, and P. W. Fowler, *Zigzags, Railroads, and Knots in Fullerenes*, 2004. doi:10.1021/ci049955h.

[9] T. Bogart, J. Goodrick, and K. Woods, *Parametric Presburger arithmetic: logic, combinatorics, and quasi-polynomial behavior*, *Discrete Analysis* (2017), article 4, 34 pp., Theorem 1.15. doi:10.19086/da.1254.

Independent researcher